

Yixiang (Henry) Wang

(347) 604-1158 | wanghenry0516@gmail.com | Pittsburgh, PA

Education

Carnegie Mellon University, Pittsburgh, PA May 2026
M.S. in Electrical & Computer Engineering |
Current Courses: Deep Learning, AI Principles, Computer Systems, Stochastic Process, Speech Recognition
Rensselaer Polytechnic Institute, Troy, NY May 2024
B.S. in Electrical Engineering | GPA: 3.99/4.0 | Magna Cum Laude |
Relevant Courses: Machine Learning, Probability Theory, Robotics

Skills

Language: Python, C, Shell Script, MATLAB
Models: LLM, Transformer, Diffusion, GNN, CNN, LSTM
Tool & Platforms: Pytorch, GitHub, Docker, Linux, Jira, Confluence, Bamboo Build, Bitbucket, Artifactory
Hardware: Circuits, Microcontrollers (MSP-EXP432P401R, Jetson Nano), Robots(Humanoid,Bionic), LIDAR

Work Experience

ASML, San Jose, CA Jun. 2024 - Aug. 2024
Software Engineering Intern

- Designed Ticket-Guided CI/CD workflow, enabling automated Bamboo builds parameterized by Jira ticket data
- Built a Ticket-Guided-Build system with Python, Shell, and REST APIs to automate software builds
- Tested the TGB system for the DevOps team through 20 trials before deploying to production
- Improved development workflow efficiency and empowered non-technical users to trigger builds
- Directed FOSS planning, establishing workflows and ensuring open-source license compliance
- Presented a FOSS implementation analysis to Brion Team, outlining timelines, phases, and stakeholders

Shenzhen Institute of AI and Robotics for Society, Shenzhen, China Apr. 2023 - Aug. 2023
Electrical Engineering Intern

- Advanced & refined path-planning algorithms in industrial robots to expedite the cargo unloading process
- Utilized Partial Integral Derivative (PID) to increase robot's accuracy and fluidity & reduce damage in mechanics
- Designed a transformable robot with pneumatic end-effectors to adapt to diverse environments through passive-active deformation
- Integrated computer vision subsystem (YOLOv8) to identify obstacles & control chamber contraction/expansion
- Designed & manufactured a control system to link the air pump & robot for control via relays & microcontrollers
- Contributed to paper writing, editing, and final publication

Rensselaer Polytechnic Institute, Troy, NY Sept. 2022 - Sept. 2023
Teaching Assistant - Electrical Engineering

- Led weekly labs and review sessions for a cohort of 30 students
- Graded 400+ assignments and provided feedback in collaboration with PhD TAs

Project Experience

Carnegie Mellon University Dec. 2024 - Feb. 2025
Team Leader

- Led a team to the top 10% in two Kaggle competitions: speech recognition and face classification/verification.
- Designed and built models from scratch, including data preprocessing, augmentation, data loaders, model architecture, loss function selection, and validation
- Conducted experiments on 8 model architectures for MLP and 5 architectures for CNN
- Conducted experiments on 5 different loss functions for fine-tuning

Chinese University of Hong Kong, Shenzhen Feb. 2023 - Apr. 2023
Research Assistant - AI-Assisted Invasive Species Detection

- Developed a database of 1,0000+ images for insect detection on imported logs, with data augmentation
- Design and built a compatible data loader for future training and fine-tuning
- Trained a residual CNN model, achieving 98% accuracy on both training and testing datasets
- Robustified the model by fine-tuning using various loss functions (ArcFace, Contrastive,Circle Loss)
- Reduced the inference time on embedded systems by 30%

Publication

J. Wang, **Y. Wang**, et al., *Transformable Inspection Robot Design and Implementation for Complex Pipeline Environment*, IEEE Robotics and Automation Letters, vol. 9, no. 6, pp. 5815-5822, June 2024. DOI: 10.1109/LRA.2024.3393208